

10 **(long)** A satellite of mass m is orbiting the earth which has radius R . The position of the satellite at any point is specified by polar coordinates (r, θ) , as shown in Fig. 10. At the perigee, the velocity of the satellite is given by $v = v_0$ and the position of the satellite is given by $(r_0, 0)$. The acceleration due to gravity at the surface of the earth is g .

(a) Obtain expressions and values for the radial and polar components of the acceleration of the satellite in the elliptical orbit. Explain why the angular momentum of the satellite is conserved about an axis passing through the centre of the earth. [6]

(b) The motion of the satellite can be expressed by the following differential equation

$$\frac{d^2}{d\theta^2}\left(\frac{1}{r}\right) + \left(\frac{1}{r}\right) = \frac{gR^2}{r_0^2 v_0^2}$$

Solve the above equation by substituting $u = 1/r$ to obtain an expression for the satellite orbit, and the eccentricity of the orbit. [8]

(c) At a given instant when the satellite is at the apogee, an impulse of magnitude I is imparted to the satellite, transferring it to a circular orbit of radius equal to the distance away from the centre of the earth at that instant. Determine the new velocity of the satellite v_1 and the magnitude of impulse I as a function of the other parameters. [8]

(d) At a later stage, following transfer to the circular orbit, the satellite commences to break up in orbit, and splits into two equal masses travelling in opposite directions. The velocity of the object travelling in the same direction as the original satellite is $3v_1$. Estimate the mechanical energy released instantaneously during satellite breakup. [8]

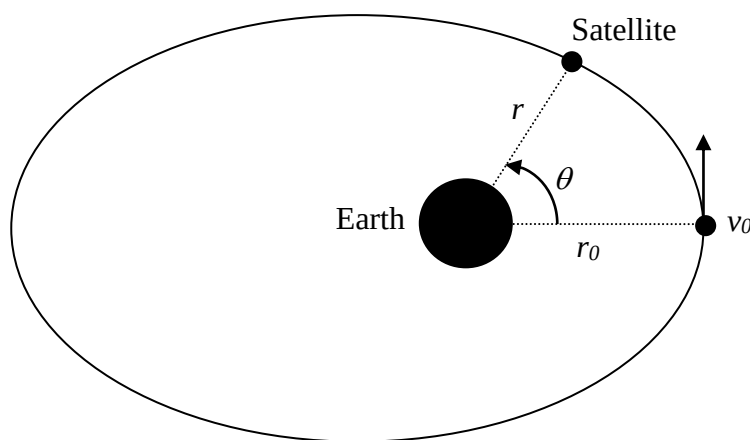


Fig. 10